Transport, Not Computation: A Cross-Model Falsification Program on How 1B–3B Language Models Use Long Context

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Abstract

We summarize a program of pre-registered, matched-control causal experiments on small open language models (Llama-3.2-1B/3B, Qwen2.5-3B) that asks a single question: when these models use long context, do they *compute* internally or merely *transport* information? The program first establishes the substrate — an exact, training-free query-axis selector and the finding that long-context retention has *no critical key budget*: the sparse compressor is not the limit, the base model is. It then runs four independent causal gates on the mechanism. **(1) Attention** up-weighting a needed key is fully explained by plain re-delivery (100% $\geq$ 80%): attention is weighted transport. **(2) Delivery ladder:** the repair for retrieval-disambiguation failure is delivering the attribute$\rightarrow$code binding — content-addressable, no positional decay, compressing to 27 bytes — while attention-redirection and transplanted state-correction are both closed; a consistent correction direction (skill score 0.46 at L8) does *not* transfer. **(3) Synthesis** of an *absent* multi-hop target reduces to retrieval-and-elimination: corrupting the middle hop flips correct implicit answers in only 5–11% of cases (vs. 84–90% for written chain-of-thought, the positive control), and blocking elimination floors implicit accuracy to chance; genuine traversal appears only as written, self-delivered CoT. **(4) Latent scratchpad:** directly *optimizing* a compact latent state (frozen weights) to switch failed chains into composition yields only answer-injection on every control — it carries the target with the code fact removed, leaves intermediates lens-invisible, is inert to hop corruption, and does not transfer — closing the door that observation alone left open. The four gates converge, cross-model and cross-family, on one thesis: *in this model class the network delivers, and composes only through an external buffer; it does not latently compute, and a latent computation state cannot be optimized into existence.* We argue the honest contribution is threefold: a map of the boundary of internal competence (a chain of well-controlled NULLs, not a new "intelligence" mechanism); a constructive, training-free engineering result (an exact long-context selector with an INT8 KV cache at 0.508$\times$ footprint, plus a 27-byte micro-hint memory architecture); and a methodology (pre-registered matched-magnitude controls that, en route, reversed two of our own false positives). All findings are regime-bounded (1B–3B, retrieval and synthetic synthesis, greedy, $N \leq$ 128K) and do not transfer up the scale; each is released as a standalone artifact with raw logs.

1 Introduction

Does a language model *think* over its context, or does it *move* pieces of it to the output? The distinction is mechanistic, not philosophical: a *computing* model forms new facts internally

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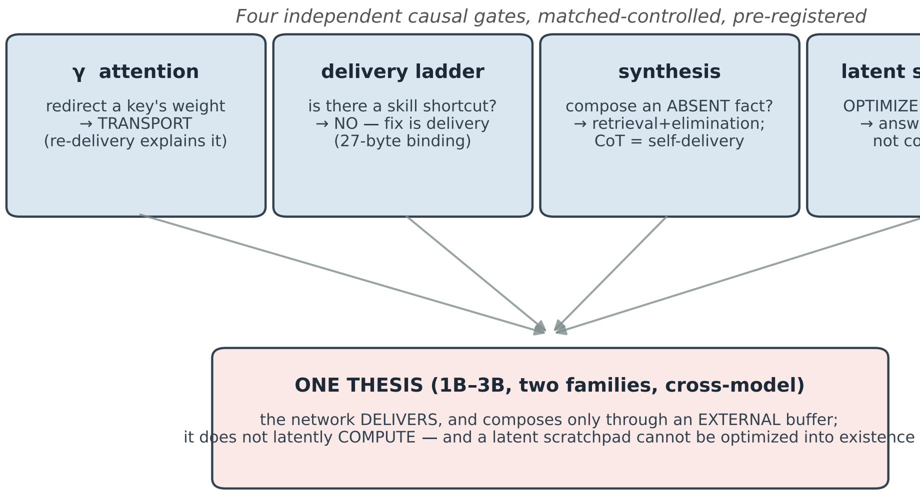

Figure 1: The program arc. Four independent, pre-registered, matched-control causal gates — on attention, on the repair of retrieval failure, on multi-hop synthesis, and on a directly-optimized latent state — converge on a single thesis, cross-model and cross-family.

(binds A, B, C into an absent D at the answer step); a *transporting* model selects, delivers, and re-reads facts that are already present, and composes — when it composes at all — by writing intermediate results back into its own context. The two are behaviourally similar on easy tasks and diverge exactly where it matters: long contexts, competition, and multi-step answers.

This paper is an umbrella over a program that pursued that distinction with one tool — the pre-registered, matched-magnitude causal gate — across small open models. The program's structure is deliberate. We first remove a confound: *is the sparse compressor itself the long-context bottleneck?* (No — §2.) With the compressor cleared, we ask, on the same models, *where does competence actually live*, through four gates that each pull a lever a system designer might reach for (§3). The gates converge (§4). We then state plainly what the program is and is not (§5), read off the engineering and product consequences (§6), and describe the methodology that makes the NULLs trustworthy, including the false positives it caught and reversed (§7).

Thesis. Across Llama-3.2-1B/3B and Qwen2.5-3B, four independent causal gates agree (Fig. 1): *the network delivers, and composes only through an external buffer; it does not latently compute.* The result is regime-bounded and is offered as a map of a boundary, not a universal law.

2 The substrate: the compressor is not the limit

Before asking how a model uses long context, we remove the obvious confound that a *sparse* engine might itself be the limiting factor.

An exact, training-free selector. Ranking keys by the projection onto the unit query direction $\mathbf{u}_Q = Q/\|Q\|$ is order-identical to ranking by the raw attention score, so top-k selection by $\langle \mathbf{u}_Q, K_i \rangle$ returns the *true* top-k with no learned parameters and no approximation [1]. The cost of truncating to k keys is governed by an attention entropy that grows logarithmically, $H_N = \alpha \log N + \beta$ with measured $\alpha \approx 0.31$, so the effective support is sublinear (N^{α}) and the perplexity gap decays as $N^{-(1-\alpha)}$ — the sparsity advantage *strengthens* with context (a descriptive scaling law; its causal status is held open).

Table 1: The four gates. Each tests a lever; each verdict is delivery/transport.

gate	decisive test	verdict
γ attention [2]	re-deliver the fact vs. up-weight its attention	**transport**: re-delivery (100%) explains the weight effect (80%); the attention-redirection class is closed
delivery ladder [3]	shrink the delivered payload; transplant a learned state correction	**delivery**: a 27-byte binding recovers 100% (content-addressable, no decay); bare code 58%; transplanted state ties random controls (not reusable)
synthesis [4]	corrupt each hop of an $A \to B \to C \to D$ chain; block elimination with decoys	**retrieval+elimination**: implicit mid-hop corruption flips 5/11% vs. 84/90% for CoT; implicit floors to chance under decoys; CoT traverses only by self-delivery
latent scratchpad [5]	optimize a latent state (frozen weights) to force composition; 5 controls	**answer-injection**: carries D with the code fact removed (0.94/0.81), intermediates stay lens-invisible, inert to hop corruption, does not transfer — no latent composition mode

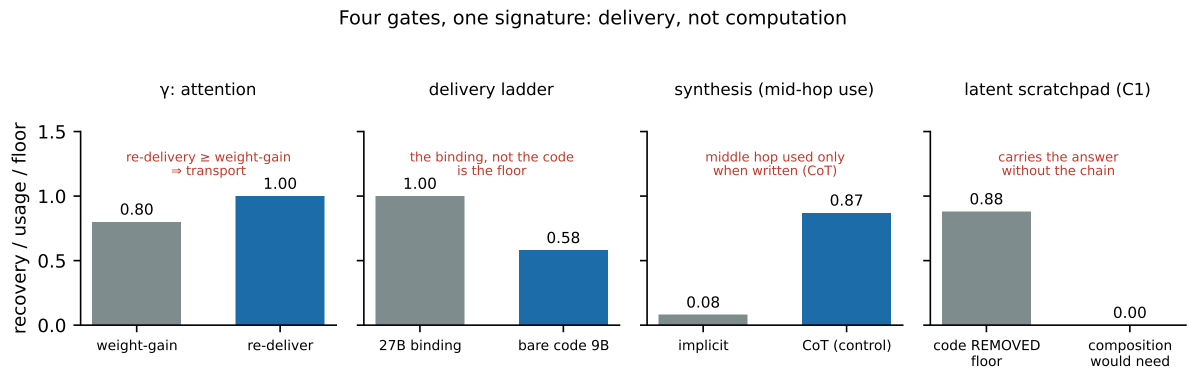

Figure 2: Four gates, one signature. In each, the contrast that would distinguish internal computation lands on the side of delivery/transport: re-delivery explains the attention effect; the binding (not the bare code) is the floor; the middle hop is used only when written down (CoT); and an optimized latent state carries the answer even with the chain's content removed.

No critical key budget. Retrieval under semantic competition has *no* absolute key budget below which it collapses: holding the budget at fixed absolute values and sweeping context length, a $4\times$ swing of the budget moves accuracy by ≤ 0.03 at any length, on three models [2]. The same budget holds at 32K yet fails at 65K; the limit is the base model's own length-driven disambiguation, not the compressor. Capacity *raises the ceiling* (the selected-yet-wrong fraction at 32K falls $0.37 \to 0.23 \to 0.18$ from 1B to 3B), and an INT8 KV cache realizes the engine at $0.508\times$ footprint for $+0.714\%$ perplexity. With the compressor cleared as the bottleneck, the mechanism question is well-posed: *the limit is in the weights — so what do the weights do with the context?*

3 Four causal gates

Each gate is pre-registered, matched-controlled, and reported with the discipline of §7. Table 1 and Fig. 2 summarize; the decisive numbers are recomputed from raw logs in each constituent paper.

γ — **attention is weighted transport.** On failure cases where the needed key is selected yet the answer is wrong, multiplying its attention logits shifts the answer strongly and needle-specifically ($\Delta P = +0.67$; a matched competitor boost gives -0.011). But plain re-delivery of the same fact recovers at least as much ($100\% \geq 80\%$): by the pre-registered rule the weight effect is fully explained by delivery. Attention here moves information; it does not compute with it.

Delivery ladder — the repair is a 27-byte binding. Re-delivering the fact recovers 100% at every distance up to 4096 tokens (content-addressable, no positional decay) and compresses losslessly to a 27-byte attribute→code binding; stripping the binding to the bare code collapses recovery to 58%. Meanwhile a learned hidden-state correction — structured enough that its cross-case consistency peaks sharply at one layer (skill score 0.46) — transplants to held-out cases no better than random controls. The repair is neither attention redirection nor stored weight-correction; it is delivery of the association.

Synthesis — "reasoning" is retrieval-and-elimination. Synthesis is the adversarial case: the target D is absent and must be assembled, so a pure transporter should fail. It does not. Implicit (no scratchpad), corrupting the middle hop flips correct answers in 5%/11% of cases while corrupting the code fact flips 100%; masking the link facts at the answer step is inert; and adding decoys that block the elimination shortcut floors implicit accuracy to chance ($0.35 \rightarrow 0.07$). The same corruption machinery detects hop usage under chain-of-thought (84/90%) — the built-in positive control proving the implicit null is real. CoT traverses genuinely, but by writing each hop into context and re-reading it (self-delivery), not by internal binding.

Latent scratchpad — composition cannot be optimized into existence. Observation has a loophole: "we did not see latent composition" is weaker than "there is none". So we search by optimization. With weights frozen, we gradient-optimize a compact latent state to make failed chains emit D; five controls separate a composition mode from a memorized answer. The optimizer succeeds at emitting D (a single virtual vector suffices) but on every control it is answer-injection: it carries D with the code fact deleted (0.94/0.81), the intermediates never rise from zero, hop corruption is inert, and it does not transfer; a CoT-informed initialization does not rescue it. Even handed maximal freedom, the reachable solution is delivery, not computation.

4 One thesis

Four levers, four independent operationalizations, two architecture families, 1B–3B: the network selects and delivers facts, composes only when it can externalize the steps into its own context, and shows no internal, decode-time formation of an absent fact — nor can such a state be optimized into being. *Transport, not computation*, within this regime. The synthesis and latent-scratchpad gates are the strongest legs precisely because they are where transport *should* have broken (an absent target; a free optimizer aimed straight at composition) and did not.

This aligns with, and sharpens, several external observations: that attention heads move information [7]; that latent multi-hop reasoning is weak and training-dependent [8, 9]; that chain-of-thought traces are not always faithful to an internal computation [10]; and that answers can be reached by elimination over the option set [11]. Our contribution is the causal, matched-control form of these readings on off-the-shelf small models, and the constructive corollary that follows.

5 What the program is, and is not

What survived (the positives). An *exact, training-free long-context selector* with a descriptive concentration scaling law and an INT8 KV cache; the finding that *retention is robust* and budget-insensitive up to where the base model itself fails; a *content-addressable delivery* result with a 27-byte micro-hint floor; and the demonstration that *capacity, not key budget*, raises the long-context ceiling.

What was closed (the NULLs). A spectral causal mechanism; an absolute critical key budget; the attention-redirection class of memory schemes; transplanted state-correction / skill-library; internal multi-hop composition at 3B; and an optimizable latent composition state. Each is a clean, controlled negative.

What is open. Whether internal composition appears *above* 3B or with trained chains; the exact channel of the above-chance implicit signal (structural/positional/elimination); and a delocalization probe (does an answer survive physical removal of the needed fact). None is closed here and none is claimed.

An honest framing. We set out hoping to find an internal mechanism of "intelligence" and instead mapped, with triple controls, the boundary of internal competence: in this regime it is delivery and externalized composition. That is the honest result — a boundary, not a breakthrough — and it is more useful than a narrative would have been. We resist the temptation to universalize: every gate is 1B–3B and regime-bounded, and a NULL at this scale does not transfer up.

6 Engineering and product corollary

The program yields a concrete, training-free architecture for a local long-context assistant, with each component justified by a gate rather than a hope:

- **Engine:** the exact query-axis selector plus the INT8 KV cache ($0.508\times$ footprint) — longer context and lower memory on consumer hardware at iso-ability. Sparsity buys speed and length, not intelligence; the safe operating coverage is well above the recency floor.
- **Memory:** store *attribute→code bindings* ($\sim$27 bytes), retrieve by attribute, deliver freshly — a RAG of micro-hints, content-addressable. No attention-steering machinery and no skill-library adapter is warranted (both closed).
- **Composition:** externalize it. Prompt the model to write its steps (CoT) or run an explicit retrieve-compose loop (self-RAG); do not expect an internal chain or a learned "thinking" vector to compose — such a vector memorizes the answer.
- **"Smarter"** on a fixed model comes from capacity or from better delivery/scaffolding, not from the sparsity budget or an internal shortcut.

7 Methodology: why the NULLs are trustworthy

Every gate fixes its grid, thresholds, controls, and verdict map *before* execution; none is tunable after. The signature move is the *matched-magnitude control*: a candidate cause is compared against a control of equal perturbation magnitude (or equal information payload), and a pre-registered null closes the direction. Verdicts are paired (within-run, immune to run-to-run nondeterminism) with bootstrap CIs, and a positive after a NULL streak is held to extra scrutiny — the discipline that matters most when one *wants* a positive.

The discipline earned its keep by catching false results: a spectral mechanism that looked causal until a matched-magnitude control reversed it; a state-transfer "signal" at 17% on n=12 that vanished at n=60; a smoke test that patched the wrong layer and produced a false zero. A separate methodological catch is independently useful: under competition, node *removal* is confounded by elimination (a removed fact is recoverable from competitors), so the faithful usage test is in-place *corruption*, not removal — a distinction that flips conclusions about which hops a chain "uses". The matched-control protocol is released as a standalone note [6].

8 Limitations

- **Scale and regime.** All gates are 1B–3B, greedy, on retrieval and synthetic multi-hop tasks, $N \leq$ 128K (synthesis N=4096). Every NULL is "not at this scale on this design", not "never"; the thesis does not transfer up.
- **Tasks.** Semantic retrieval-under-competition and synthetic attribute→person→zone→code chains; natural reasoning, planning, and code are separate gates.
- **The descriptive scaling exponent** α is single-model/corpus and its causal status is held open; the full multi-seed PPL sweep is a separate artifact.
- **Off-distribution interventions** (attention forcing, mask-at-answer, latent optimization) are off-distribution by construction; degeneration was distinguished from clean answer-death, and "optimization always finds something" was defused by controls, not assumed away.

9 Conclusion

A program of pre-registered, matched-control causal gates, cross-model over two families and 1B–3B, converges on one regime-bounded thesis: these models *transport* information and compose only through an external buffer; they do not latently compute, and a latent computation state cannot be optimized into existence. The honest yield is a map of the boundary of internal competence, a training-free long-context engine and a 27-byte micro-hint memory architecture justified by that map, and a falsification methodology that reversed its own false positives. Whether internal composition lives above this scale is the next program, and it should be approached with the same matched-control skepticism that produced these results.

Program artifacts

Each finding is a standalone release with raw logs, pre-registration, and a figure script that recomputes its numbers. (1) Exact ranking + entropy scaling [1]; (2) No critical key budget [2] (DOI 10.5281/zenodo.20514229); (3) The bottleneck is delivery [3]; (4) Even synthesis is transport [4]; (5) No latent scratchpad [5]; (6) Matched-magnitude controls [6] (DOI 10.5281/zenodo.20261207). This umbrella references their verified headline numbers; the constituent repositories hold the data and reproductions.

References

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